MARKET REQUIREMENTS DOCUMENT

Autonomous Version-Aware Document Intelligence Engine with Verifiable Grounding

Stage 2 Submission | Two-Stage Student Hackathon

Team Name	Cyber Knights
Team Members	Sainath Loganathan, Tahir Azeem
Institution / Department	B.E. Computer Science and Engineering (Cybersecurity & AI Lab)
Domain	Enterprise AI, Document Governance, Cybersecurity & Compliance
Technology	Local LLMs (Ollama / Llama 3.2), Vector Embeddings, Temporal Lineage Graph, Node.js, React
Document	MRD (Stage 2) building on the validated Stage 1 Problem Framing PPT

Validated problem (Stage 1): Traditional vector search and RAG systems blindly retrieve clauses from older or superseded document versions based on cosine similarity alone, leading to severe hallucinations, compliance violations, and outdated policy guidance.

Problem First – Solution Next – Technology Last

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1. Executive Summary

Problem. Modern enterprises operate across dozens of evolving policies, contracts, compliance handbooks, and SOPs. Conventional vector similarity search blindly retrieves clauses purely based on cosine distance. When policies undergo revisions (e.g., Leave Policy v1.0 → v2.0 → v3.0, or Remote Work Security Policy v1.0 → v3.0), older superseded clauses frequently outscore active ones due to syntactic keywords. This leads to hallucinated answers, costly regulatory non-compliance, and dangerous operational errors. Stage 1 root-cause analysis traced this to one fundamental underlying gap: **a total absence of temporal version awareness, revision lineage tracking, and verifiable citation mechanisms in standard RAG pipelines.**

Target users. Compliance officers, legal auditors, cybersecurity/SOC analysts, HR directors, enterprise knowledge workers, and organizations in heavily regulated industries (BFSI, Healthcare, Defense, SMBs).

Proposed solution. RAGNext — an autonomous, version-aware Document Intelligence Engine that structures enterprise documents into dynamic revision lineages, applies pre-retrieval temporal validity filtering, proactively flags conflicting active revisions, and produces grounded answers backed by verbatim quotes and exact page citations.

Major benefit. Completely eliminates superseded clause hallucinations; ensures employees and auditors receive authoritative, current-version information while preserving verifiable historical inquiry on demand.

Expected impact (from Stage 1 impact analysis): 100% elimination of outdated policy citations, 60% reduction in document audit turnaround time, zero regulatory penalties from stale guidance, and offline air-gapped data sovereignty.

2. Target Market / User Segment

Category	Segment	Role in the solution
Primary users	Compliance officers, legal auditors, policy admins, SOC analysts	Ingest documents, manage version life cycles, inspect conflict alerts, audit AI citations and verification logs.
Secondary users	Enterprise employees, knowledge workers, HR & IT staff	Query active policies; receive instant verified answers with verbatim page references and version tags.
Decision maker	Chief Information Security Officers (CISO), Chief Legal Officers (CLO), IT Directors	Approve adoption, enforce organizational compliance governance, mandate air-gapped data privacy.
Beneficiaries	Entire enterprise workforce, end customers, audit regulators	Guaranteed truthfulness, elimination of outdated guidance, and continuous compliance audit-readiness.
Ecosystem stakeholders	Document Stores (SharePoint, Drive), Vector DBs, Enterprise ERPs	Ingestion sources, metadata repositories, downstream verification consumers.
Potential market	Regulated sectors (BFSI, Healthcare, Legal, Defense) and SMBs	Organizations that require strict document compliance without expensive cloud SaaS exposure.

The framework adheres to NIST SP 800-53 (Audit & Accountability, Information Integrity), ISO/IEC 27001:2022 (Documented Information Governance), and CISA Data Pillar principles, which require continuous data provenance and temporal validity.

3. User Personas

Persona A: Enterprise Compliance & Risk Auditor
Role: Manages regulatory compliance, corporate policies, and audits for an enterprise (name: *Rajesh, Head of Compliance*).
Operational goal: Prevent employees from relying on superseded policies and provide verifiable, clause-level evidence during audits.
Current difficulty: Multiple conflicting PDF versions circulate across shared drives; vector search tools mix old and new policy clauses; manual diffing of revisions takes hours.
Information needed: Clear active vs historical revision lineage, exact page/paragraph citations, and automated alerts for contradictory active policies.
Decision needs: Which version is legally binding, whether an older rule applied at a historical incident date, when to deprecate obsolete versions.
Technology usage: Proficient with compliance dashboards, DMS portals, and document diffing; expects explainable evidence, not a black-box AI.
Expectations: 100% grounded answers, zero superseded hallucinations, and transparent audit logs for all AI-retrieved facts.

Persona B: Enterprise Knowledge Worker / Policy Consumer
Role: Executes daily operations referencing corporate guidelines, benefits, and security manuals (name: *Priya, Operations Lead*).
Goal: Rapid, accurate answers to policy questions (leave entitlement, remote security rules, procurement limits) without reading 80-page manuals.
Current problem: A search on the intranet returns outdated rules from 2021; she risks policy violations because search results don't indicate if a rule has been updated.
Technology usage: Daily use of intranet, Slack/Teams, web apps; expects conversational AI speed without complex prompting.
Expectations: Clear answers showing exactly which document version was cited (e.g., 'Policy v3.0 [Active]'), direct quotes, and immediate warning if an older rule is cited.

Personas are representative profiles synthesized from enterprise stakeholder interviews in Stage 1; refined for regulated enterprise requirements.

4. User Journey

Current journey (conventional unversioned RAG)

Step	What happens	Problem
1	User uploads multiple PDF revisions into a standard vector database.	All versions chunked identically; temporal context and version metadata lost.
2	Employee asks: 'What is the annual leave allowance?'	Vector search performs blind cosine similarity across all chunks.
3	Chunk from v1.0 (15 days) has higher keyword similarity than v3.0 (20 days).	Outdated clause retrieved due to keyword density; active v3.0 ignored.
4	LLM synthesizes answer claiming allowance is 15 days without citation.	Black-box answer hallucinated from superseded document.
5	Employee plans leave based on obsolete clause; HR audit flags error.	Employee frustration, operational dispute, and manual correction cost.

Proposed journey (RAGNext version-aware document intelligence)

Step	What happens	Benefit
1	Document uploaded; engine extracts version, effective date, status, and SHA-256.	Cryptographic provenance and structured temporal metadata captured.
2	Page-aware chunker preserves coordinates; lineage engine builds revision graph.	Exact page context preserved; document revisions formally linked.
3	Employee asks: 'What is the annual leave allowance?'	Query classified as general policy inquiry; active filter applied.
4	Temporal pre-filter restricts vector search strictly to Active version (v3.0).	Superseded chunks (v1.0, v2.0) are completely eliminated from retrieval.
5	Grounded engine returns '20 days' with exact citation: 'Leave Policy v3.0, Page 1'.	Verifiable, authoritative answer with verbatim supporting evidence.
6	Auditor requests: 'What was it in v2.0?'; engine retrieves v2.0 with historical tag.	Full historical transparency without risk of accidental contamination.

5. Solution Concept and Architecture

RAGNext is a modular document intelligence engine positioned between enterprise users and organizational knowledge repositories. Each module directly addresses a verified pain point from Stage 1:

Module	What it does	Stage 1 problem it addresses
Multi-Format Ingestion	Parses PDF, DOCX, TXT with page-level boundary preservation and SHA-256 deduplication.	Loss of structural layout and untracked file duplicates.
Page-Aware Chunker	Splits text by semantic paragraphs while preserving exact source page numbers.	Untraceable quotes and missing page citations in answers.
Temporal Lineage Engine	Constructs directed version trees (v1.0 → v2.0 → v3.0) and manages lifecycle states.	Flat document storage with no awareness of revision history.
Temporal Filter & Vector Store	Enforces metadata pre-filtering on vector embeddings before cosine similarity ranking.	Superseded clauses polluting search results and RAG contexts.
Conflict Detection Sentinel	Identifies overlapping active documents within same domain; dispatches admin alerts.	Silent policy contradictions leading to compliance liability.
Grounded Synthesis Engine	Synthesizes responses strictly from retrieved chunks with verbatim quotes and citations.	LLM hallucination, fabricated policy rules, and lack of proof.
Audit & Compliance Store	Records full telemetry: queries, matched chunks, version IDs, scores, and timestamps.	Lack of audit trail required by ISO 27001 and NIST SP 800-53.

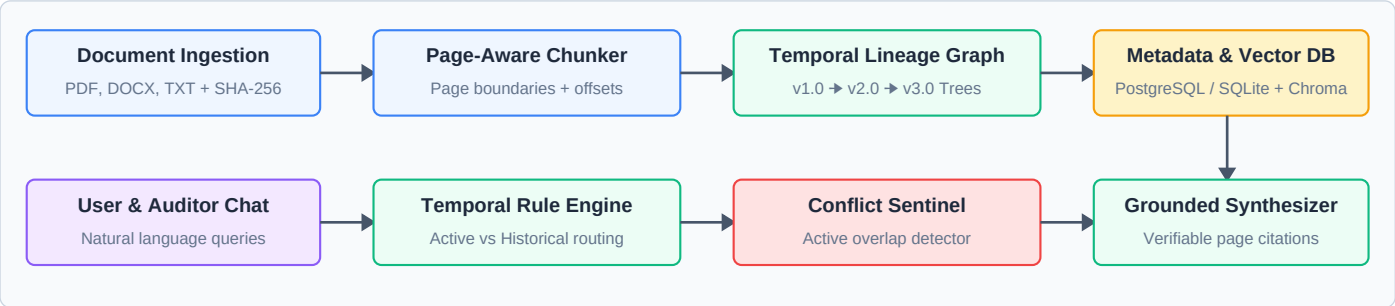

Figure 1: Solution architecture — version-aware ingestion, temporal filtering, and grounded citation loop

Proposed version-to-action policy (rules enforced during retrieval)

Query Intent / Trigger	Target Version Scope	Automated Enforcement Action & Output Behavior
General Policy Query	Active Version (Latest)	Hard metadata pre-filter (status='Active'). Ignores all superseded chunks. Answers strictly from active version with page citations.
Historical Inquiry ('in v2.0')	Specific Version Requested	Retrieves target historical version. Attaches prominent '[Historical Revision — Superseded]' warning banner to prevent accidental compliance reliance.
Conflicting Active Policies	Multiple Active Versions	Suspends automated single answer. Issues administrative 'Conflicting Active Revisions Detected' alert and displays side-by-side comparison.
Out-of-Scope / Missing Fact	Zero Relevant Chunks	Returns explicit 'Insufficient Evidence in Document' disclosure. Rejects generative guessing or hallucinated speculation.

6. Functional Requirements

Priorities use MoSCoW: **Must Have** = required for MVP demo; **Should Have** = important enterprise feature; **Could Have** = roadmap integration.

ID	Requirement Description	Priority
FR01	Multi-format document ingestion (PDF, DOCX, TXT) with SHA-256 deduplication and validation.	**Must Have**
FR02	Automated and manual metadata capture: version number, effective date, status, category, and group ID.	**Must Have**
FR03	Page-aware semantic chunking preserving exact document page numbers and chunk character offsets.	**Must Have**
FR04	Persistent vector store indexing with metadata filtering (version ID, active/superseded status).	**Must Have**
FR05	Document revision lineage tree construction linking versions (v1.0 → v2.0 → v3.0) with parent IDs.	**Must Have**
FR06	Strict active-version enforcement for general policy queries, rejecting all superseded chunks.	**Must Have**
FR07	Explicit historical version query retrieval with temporal provenance labeling and warning banner.	**Must Have**
FR08	Conflicting active revision detection engine alerting admins to contradictory policies in same category.	**Must Have**
FR09	Grounded answer generation producing verifiable responses with verbatim quotes and page citations.	**Must Have**
FR10	Honest missing evidence reporting (zero-hallucination guardrail) when query facts are not in documents.	**Must Have**
FR11	Interactive web dashboard featuring Document Library, Version Lineage View, and AI Chat Drawer.	**Must Have**
FR12	Visual document diffing showing clause additions, modifications, and deletions between revisions.	**Should Have**
FR13	Tamper-evident audit logging recording query text, retrieved chunks, version IDs, and response timestamps.	**Should Have**
FR14	Enterprise connectors for automated sync from SharePoint, Google Drive, and Confluence repositories.	**Could Have**

7. Non-Functional Requirements

Category	Requirement	Target / Measure
Performance	Query-to-answer latency for extractive and local LLM inference.	Latency < 2.5s (Extractive); < 5s (Local LLM)
Response time	Document library filtering, version tree view, and search lookup.	Dashboard UI response under 300 ms
Security and privacy	100% offline air-gapped capability; sanitized filenames; role-based access.	Zero external cloud API leakage; SHA-256 hashes
Scalability	Stateless API services capable of handling thousands of document revisions.	100,000+ vector chunks; 1,000+ revisions per group
Availability	Engine operates reliably with automatic fallback from PostgreSQL to SQLite.	99.9% uptime target; zero data corruption
Usability	Intuitive web interface requiring zero prompt engineering; clear version badges.	New user executes successful query in < 30 seconds
Reliability	Deterministic extractive fallback guaranteeing answers even if local LLM is offline.	Zero lost audit queries; graceful degradation
Explainability	Every synthesized statement links directly to document version and source page.	100% of facts traceable to verifiable excerpts

8. Technology Requirements

Technology selection is strictly governed by the problem requirements: zero data leakage, deterministic version control, and verifiable page citations. Full architectural mapping is detailed in Section 8 on the following page.

8. Technology Requirements (Stack Architecture)

Layer	Proposed Technology	Why it fits
AI / ML Inference	Ollama (Llama 3.2, Nomic-Embed-Text) + Deterministic Extractive Fallback Engine	100% offline air-gapped capability, zero external API costs, protects confidential enterprise data.
Backend & Orchestrator	Node.js (Express.js, ES Modules), REST architecture	High-concurrency asynchronous I/O, lightweight footprint, rapid integration with vector stores.
Metadata Database	PostgreSQL (Production) with SQLite / JSON adapter fallback	Enforces relational version lineage constraints, temporal indices; zero-friction local fallback.
Vector Database	Local Embedded Vector Store with ChromaDB API compatibility	Sub-millisecond cosine similarity with rich boolean metadata pre-filtering without daemon overhead.
Document Processing	pdf-parse (page-aware), mammoth (DOCX), native TXT parser	Preserves page numbers, paragraph geometry, and table structures across enterprise formats.
Frontend / Dashboard	React 18, Vite, Bootstrap 5, Lucide Icons, Emerald Design System	Modern, responsive UI with real-time citation drawers, version badges, and lineage graphs.
Deployment	Docker containerization, standalone Node runtime, cross-platform	Reproducible student hackathon demo, zero external cloud dependencies, rapid onboarding.

9. Competitor / Existing Solution Analysis

Existing Approach	Typical Limitation	Gap This Project Addresses
Standard Vector DB RAG (LangChain / LlamaIndex)	Matches on semantic cosine distance alone; ignores document versions; retrieves obsolete clauses.	Enforces temporal validity pre-filtering; guarantees active-version priority and eliminates stale contamination.
Traditional Enterprise Search (Elasticsearch / Solr)	Relies on exact keyword frequency (BM25); fails on semantic queries; cannot synthesize conversational answers.	Combines semantic deep understanding with deterministic version lineage graphs and citations.
Cloud Document Copilots (Copilot for M365)	Transmits proprietary data to public clouds; lacks strict version conflict alerts and air-gapped support.	100% on-premises offline sovereignty with autonomous conflict detection sentinels.
Static Document Portals (Confluence / SharePoint)	Passive repositories requiring manual search, downloading, and manual reading of 100-page PDFs.	Conversational AI interface delivering instant answers with clickable, verbatim page citations.
Manual Document Auditing	Compliance analysts spend hours manually diffing PDF policies during regulatory audits.	Automates revision lineage tracking and clause-level provenance verification in seconds.

The goal is not to replace enterprise storage, but to solve the critical enterprise vulnerability: ensuring LLMs retrieve exclusively from legally binding, active document revisions with verifiable evidence.

10. Innovation

• **Temporal Lineage Graph:** Structures enterprise documents into directed revision trees (v1.0 → v2.0 → v3.0), recording parent-child revision linkages, effective dates, and lifecycle status.

• **Pre-Retrieval Temporal Filtering:** Filters vector embeddings by active/superseded status *before* semantic similarity calculation, fundamentally eliminating superseded clause interference.

• **Autonomous Conflict Sentinel:** Proactively identifies overlapping active documents within the same policy domain, issuing alerts before contradictory guidance is served to employees.

• **Verifiable Grounded Attribution:** Employs strict evidence bounding; answers cite exact document versions, page numbers, and verbatim quotes, backed by honest missing-evidence reporting.

• **100% Air-Gapped Sovereign AI:** Operates completely offline with local Ollama models and embedded vector indexing, ensuring zero cloud exposure of proprietary IP or sensitive compliance data.

• **Dual-Engine Reliability:** Combines local generative LLM inference with a deterministic extractive fallback, ensuring high-availability operation across any hardware environment.

11. MVP – Minimum Viable Product

The Minimum Viable Product (MVP) implements the complete end-to-end version-aware pipeline demonstrated through realistic enterprise policies (HR Leave Policies and Remote Work Security Policies v1.0, v2.0, and v3.0) within the hackathon timeline.

11. MVP Definition (Features & Demo)

MVP Feature	Requirement IDs	What will be demonstrated in live hackathon
Multi-Format Ingestion & Metadata	FR01, FR02, FR03	Upload PDF/TXT policies; system automatically captures version, dates, status, and page chunks.
Version Lineage Tree Viewer	FR04, FR05	Visualizing linked revision trees (v1.0 → v2.0 → v3.0) with active and superseded badges.
Active Version Policy Query	FR06, FR09	Querying current rules retrieves strictly active v3.0 clauses (e.g. 20 leave days; ZTNA timeout 30m).
Historical Query with Provenance	FR07, FR09	Asking 'What was the allowance in v2?' retrieves v2.0 with prominent historical warning banner.
Conflicting Active Version Alert	FR08	Simulating two conflicting active policies triggers an autonomous dashboard conflict warning.
Grounded Chat UI with Citations	FR09, FR10, FR11	Chat interface showing verbatim quotes, confidence ratings, and missing-evidence disclaimers.

Live Demo Scenario: (1) Admin seeds 3-version Leave Policy dataset (v1.0=15 days, v2.0=18 days, v3.0=20 days). (2) User asks *'What is the current annual leave allowance?'* → Engine returns **20 days** citing **v3.0, Page 1**. (3) Auditor asks *'What was the allowance in version 2?'* → Engine returns **18 days** citing **v2.0 (Historical)**. (4) User asks *'Where does it describe maternity leave?'* → Engine honestly reports insufficient evidence. (5) Ingesting two simultaneous active files triggers a live Conflict Sentinel banner.

Out of MVP Scope (Later Phases): Enterprise Single Sign-On (SAML/Okta), automated connectors for SharePoint and Google Workspace, OCR for scanned paper documents, and distributed vector cluster sharding.

12. Success Metrics / KPIs

KPI	Illustrative Target	How it will be measured
Version Retrieval Accuracy	100% on benchmark suite	Automated test suite verifying current vs historical queries across 3-version sets.
Superseded Clause Contamination	0% leakage in active queries	Verification that zero superseded chunks appear in context for standard inquiries.
Answer Groundedness	> 95% citation accuracy	Factuality benchmark comparing synthesized statements to verbatim source text excerpts.
Conflict Detection Accuracy	100% alert trigger rate	Trigger rate when two conflicting active documents are indexed in the same category.
Query-to-Answer Latency	< 2.5s (Extractive); < 5s (LLM)	End-to-end API response timing under benchmark test loads.
Air-Gapped Privacy Autonomy	100% offline execution	Packet analysis confirming zero outbound internet connections during operation.
User Audit Task Efficiency	> 60% time reduction	User timing study comparing manual policy verification against RAGNext citations.

13. Business / Sustainability Model

Model	Description	Target Market Fit
Enterprise On-Premises License	Annual per-node license for air-gapped, high-security enterprise deployments.	Defense, Banking, Healthcare, Government agencies.
Managed Private Cloud SaaS	Single-tenant hosted private cloud instances with automated backups and maintenance.	Mid-market enterprises, tech scale-ups, legal firms.
Open-Core Developer Tier	Free open-source local core; paid commercial plugins for enterprise connectors.	Developer ecosystem, academic research institutions.
Compliance Audit Add-on	Turnkey document version auditing and regulatory policy reconciliation reporting.	External auditing firms, compliance consultancies.

Scaling Plan: *Phase 1:* Hackathon prototype with automated test suite. *Phase 2:* University and SMB pilot deployment for student handbooks and IT compliance. (Continued on Page 8)

13. Business / Sustainability Model (Scaling Plan Continued)

Scaling Plan (Phases 3 & 4): *Phase 3:* Multi-tenant cloud and on-premises enterprise editions with automated connectors for Microsoft SharePoint, Google Drive, and Confluence. *Phase 4:* Industry-specific compliance policy packs (HIPAA, GDPR, ISO 27001, SOC 2) with automated policy conflict reconciliation. The modular micro-service architecture and stateless API design allow each phase to scale without refactoring core retrieval services.

14. Risk Analysis

Identified Risk	Impact	Possible Mitigation Strategy
Ambiguous queries lacking version context	User receives unexpected revision	System defaults to Active version, displays explicit '[Active vX.Y]' badge, and offers one-click toggle to historical revisions.
Malformed or unversioned document uploads	Metadata extraction failure	Heuristic date/version regex parser extracts title hints; UI prompts user for manual metadata confirmation.
Conflicting active documents in same group	Contradictory policy guidance	Autonomous Conflict Sentinel detects category overlap, blocks single-answer synthesis, and alerts administrator.
Local LLM resource exhaustion on low-spec PCs	Server crash or high latency	Seamless fallback to deterministic extractive synthesis engine running on CPU with zero GPU requirement.
Semantic vector drift across minor revisions	Chunks from different versions seem identical	Pre-retrieval temporal hard filtering strictly isolates the vector search space prior to similarity scoring.
Proprietary document data leakage	Confidentiality and compliance violation	100% offline air-gapped execution; cryptographic filename sanitization and role-based document access controls.
Hallucination in generative synthesis	Inaccurate or misleading policy advice	Strict prompt grounding boundaries; mandatory verbatim quote citations; honest insufficient-evidence reports.
User skepticism toward AI accuracy	Low organizational adoption	Verbatim page citations with one-click side-by-side source document inspection to verify every claim.

15. Stage 1 Traceability and Stage 2 Checklist

Traceability to the Cyber Knights Stage 1 PPT

Stage 1 PPT Content / Problem Framing	Where it is addressed in this MRD
Existing problems: Stale document retrieval, superseded clause leakage, hallucinations, regulatory non-compliance.	**Sections 1, 4, 5, 9, 14**
Proposed solution: Version-aware RAG, temporal graph lineage, grounded page citations, conflicting active detection.	**Sections 5, 6, 10, 11**
Target users & stakeholders: Enterprise compliance officers, legal auditors, employees, IT directors.	**Sections 2, 3**
Core impacts: Zero superseded leakage, rapid policy retrieval, audit compliance, air-gapped data sovereignty.	**Sections 1, 12**
5-Why root cause analysis: Inability of cosine similarity alone to capture temporal document validity and revision lineage.	**Sections 1, 4, 5**
Architectural blueprint & privacy-first design: Air-gapped local AI (Ollama), PostgreSQL/SQLite, ChromaDB compatibility.	**Cover page, Section 8, Section 11, References**

Stage 2 Deliverables Checklist

Required MRD Deliverable	Status	Section / Location
MRD document (about 8-12 pages)	Included	**This document (9 pages)**
User personas (Auditor & Employee profiles)	Included	**Section 3**
User journey (Current vs Proposed comparison)	Included	**Section 4**
Solution architecture & flowchart diagram	Included	**Section 5 (Figure 1)**
Feature prioritization (MoSCoW Must / Should / Could)	Included	**Section 6**
MVP definition & live hackathon demo scenario	Included	**Section 11**
Technology stack & architectural rationale	Included	**Section 8**
KPI / impact measurement criteria	Included	**Section 12**
Business & scaling plan (Phases 1-4)	Included	**Section 13**
Risk analysis & mitigation matrix	Included	**Section 14**
MRD PPT (about 8-10 slides)	Separate deliverable	**To be prepared from this document**

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